

Final Project Proposal

For the final project, I would like to study a dataset from the National Institute of Diabetes and Digestive and Kidney Diseases. The dataset contains patient history and biometric data from 768 women aged 21 or older, including blood pressure, glucose levels, insulin levels, and the presence of diabetes. The dataset can be found at: kaggle.com/datasets/mathchi/diabetes-data-set/data.

The primary goal of this project will be to understand the relationship between the available patient information and the occurrence of diabetes. Because diabetes is a widely prevalent and impactful health condition in the U.S. and around the globe, it is extremely important to study the effect of different biological factors on one's risk for diabetes. An understanding of how certain variables in patient history are related to the condition may help identify signals which can predict one's risk of diabetes, to enable more effective treatments and preventative measures.

My current plan for the project has three components:

- 1) Ideally, I would like to create a logistic regression model to predict the presence of diabetes in a patient given their biometric data, and fit the parameters of the model using Bayesian inference (as long as this fits the scope of the course). A linear regression model for glucose levels may be a suitable alternative.
- 2) In addition, I plan to study more simply how the distributions for blood pressure, glucose levels, insulin levels, age, and number of pregnancies differ across those with diabetes and those without. For example, I may build a Poisson model for the number of pregnancies of patients with and without diabetes, and then compare and contrast features of their posterior distributions. Similarly, I may build a multivariate normal model for BMI, blood pressure, glucose, and/or insulin levels.
- 3) For the purpose of building a multivariate normal model (or for general knowledge), I would like to estimate the correlations between these variables using Bayesian inference.

Fitting the previously mentioned normal model will likely require the use of MCMC methods. In addition, due to the fact that the sample distributions for insulin and glucose levels are significantly skewed, MCMC may be especially warranted due to a potentially complex sampling model and resulting posterior.